

Digvijay Singh Shekhawat

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Summary

Artificial Intelligence and Machine Learning undergraduate with hands-on experience developing AI-powered data analytics applications, machine learning pipelines, and deep learning solutions. Experienced with Python, SQL, Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), computer vision, and predictive analytics. Passionate about building end-to-end AI systems that transform structured and unstructured data into actionable business insights.

Education

SOA - Institute of Technical Education and Research, Btech in Computer Science (Data Science Specialization) 2022-2026

- GPA: 7.5/10.0

Experience

Anmoul Infomatics, ML intern Aug 2025 – Jul 2026

- Automated cleaning, validation, and preprocessing of structured Excel/CSV datasets using Python, enabling reliable exploratory data analysis and visual reporting for downstream stakeholders.
- Built a document processing pipeline using Tesseract OCR to extract text and structured fields from scanned PDFs and multi-format files, forming the ingestion layer for downstream NLP tasks.
- Applied NLP and machine learning techniques for document classification and information extraction, converting unstructured text into structured, queryable fields.
- Engineered a semantic search system by generating document embeddings and applying similarity-based retrieval, cutting manual document lookup effort by enabling instant relevant-result retrieval.
- Designed and implemented an LLM-based knowledge assistant using a Retrieval-Augmented Generation (RAG) architecture, iterating on document chunking strategy and retrieval logic to improve response relevance and grounding.

Technical Skills

Languages: Python, SQL, Java, C, C++

Machine Learning & Deep Learning: Scikit-Learn, PyTorch, TensorFlow, XGBoost, Gradient Boosting, CNNs, Vision Transformers, Transfer Learning, Fine-Tuning, AutoML, Feature Engineering, Model Evaluation

Generative AI & NLP: LLMs, Hugging Face Transformers, LangChain, LlamaIndex, RAG, Sentence Transformers, Embeddings, Semantic Search, Prompt Engineering, AI Agents, Function Calling, Tool Calling

Data Engineering: PostgreSQL, Apache Spark, PySpark, ETL, Data Pipelines, Data Cleaning, Data Wrangling, Data Validation

Development & Visualization: FastAPI, Streamlit, Gradio, Git, GitHub, Plotly, Matplotlib, Seaborn

Projects

NLP-Based Automated Fake News Detection GitHub

- Built end-to-end fake news classifier on the LIAR2 dataset (18,369 train / 2,296 test political statements), combining text features with speaker credibility history.
- Fine-tuned bert-base-uncased with a hybrid classifier head fusing CLS embeddings with credibility features, achieving 0.883 weighted F1, outperforming a traditional SVM baseline (0.868 F1).
- Engineered a TF-IDF pipeline (50,000 features) and benchmarked SVM, Random Forest, and BiLSTM baselines to validate model selection trade-offs between accuracy and compute cost.
- Applied SHAP to the SVM model across 200 test samples, generating feature-importance explanations of linguistic and credibility signals driving predictions.

Cricket Fantasy Point System

Live Demo

- Built a predictive fantasy scoring system using historical ball-by-ball IPL data, achieving approximately 65% prediction accuracy when evaluated against historical data.
- Engineered player performance features and applied AutoML (PyCaret) with Gradient Boosting to generate optimized Dream11 team recommendations.

DocXtract – Document OCR & Structured Data Extraction

Live Demo

- Built an end-to-end OCR pipeline using Tesseract to extract text from scanned documents and PDFs across multiple formats (PNG, JPG, TIFF, PDF).
- Designed a keyword-based document classification engine to automatically identify document type (driving license, passport, W-2, paystub, flood certificate) from OCR output.
- Implemented document-specific regex and rule-based extraction logic to convert unstructured OCR text into structured JSON fields, tailored per document type.

PneumoVision-ViT: Pneumonia Detection from Chest X-Rays

Live Demo

- Developed a Vision Transformer (ViT) model for pneumonia detection from chest X-ray images, achieving 75.64% test accuracy and an 83.55% F1-score.
- Applied transfer learning, fine-tuning, and data augmentation to improve model generalization on medical imaging data.

Key Achievements

FIFS Sports Data Gameathon 2025 1.0: Qualified for the final round and secured a Top 20 finish among national participants.

FIFS Sports Data Gameathon 2025 2.0: Achieved 1st Place (Winner) among top university teams in a national-level competition. *Certificate*

Amazon ML Challenge 2024: Ranked in the Top 500 teams out of approximately 10,000 teams on the public leaderboard.

Certifications

IBM Data Analyst Professional Certificate — Coursera

Data Science using Python — Coursera

Google AI Essentials

Hugging Face NLP Course

Kaggle: Python, Pandas, Machine Learning, Feature Engineering & Deep Learning